Chapter 12 - Magnetic Effect of Electric Current



A magnet is a object which attracts pieces of iron, nickel and cobalt.

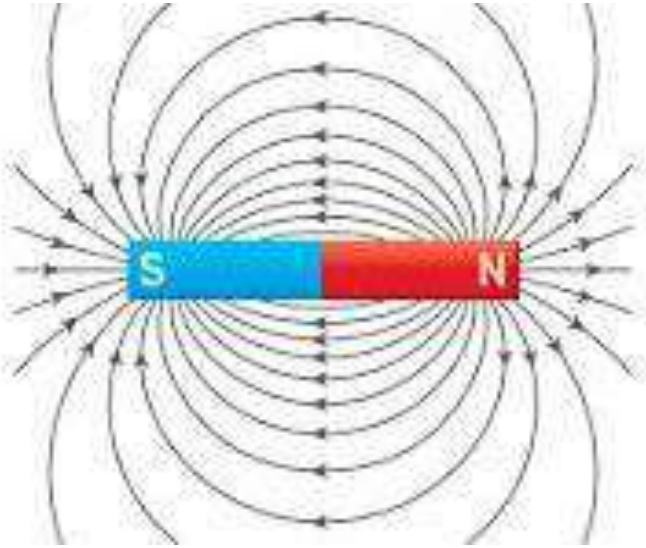
A bar magnet

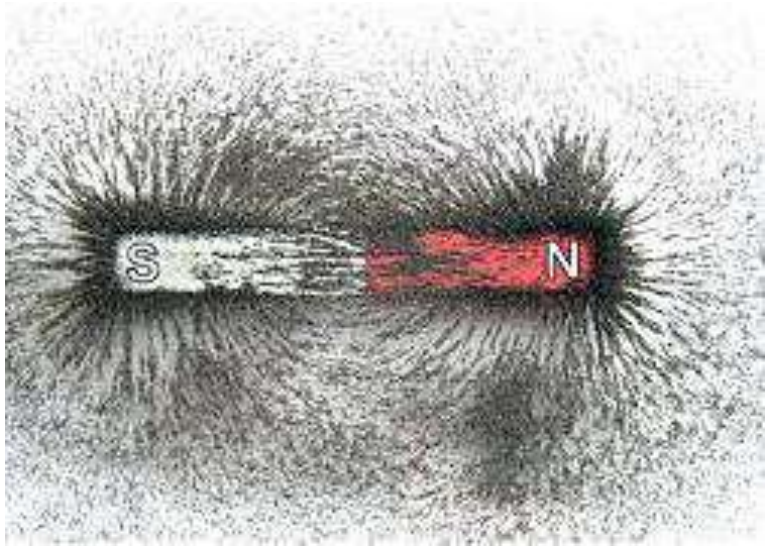
A **bar magnet** is a rectangular block that acts like a invisible magic stick, pulling things made of iron right toward it. It has two ends called the North and South poles, where opposite ends stick together while matching ends push each other away.



Magnetic Field

- A **magnetic field** is the invisible space or area around a magnet where its magnetic force works. If an object made of iron or another magnet enters this area, the magnet can pull or push it without even touching it.

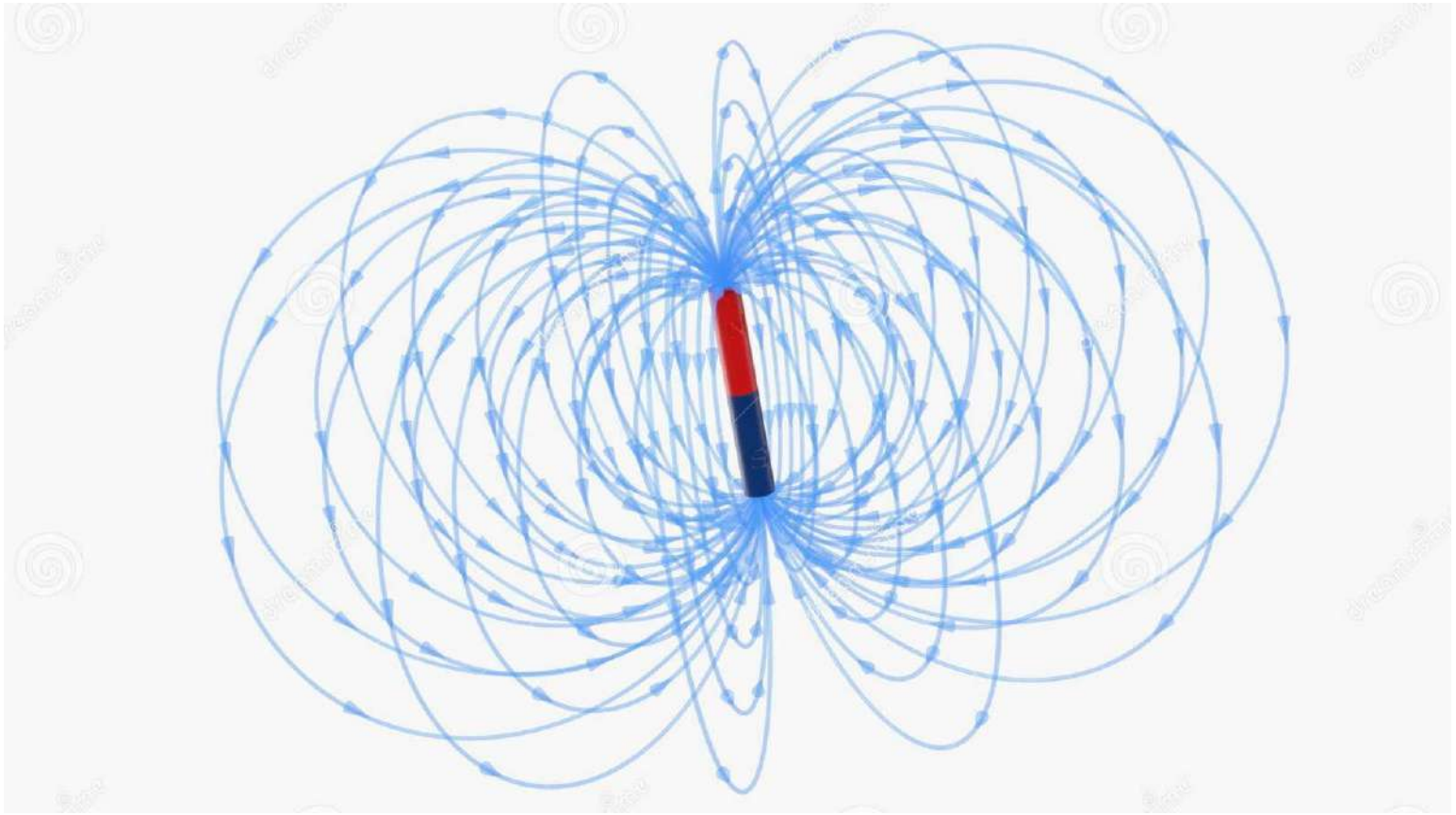




Properties of magnetic field

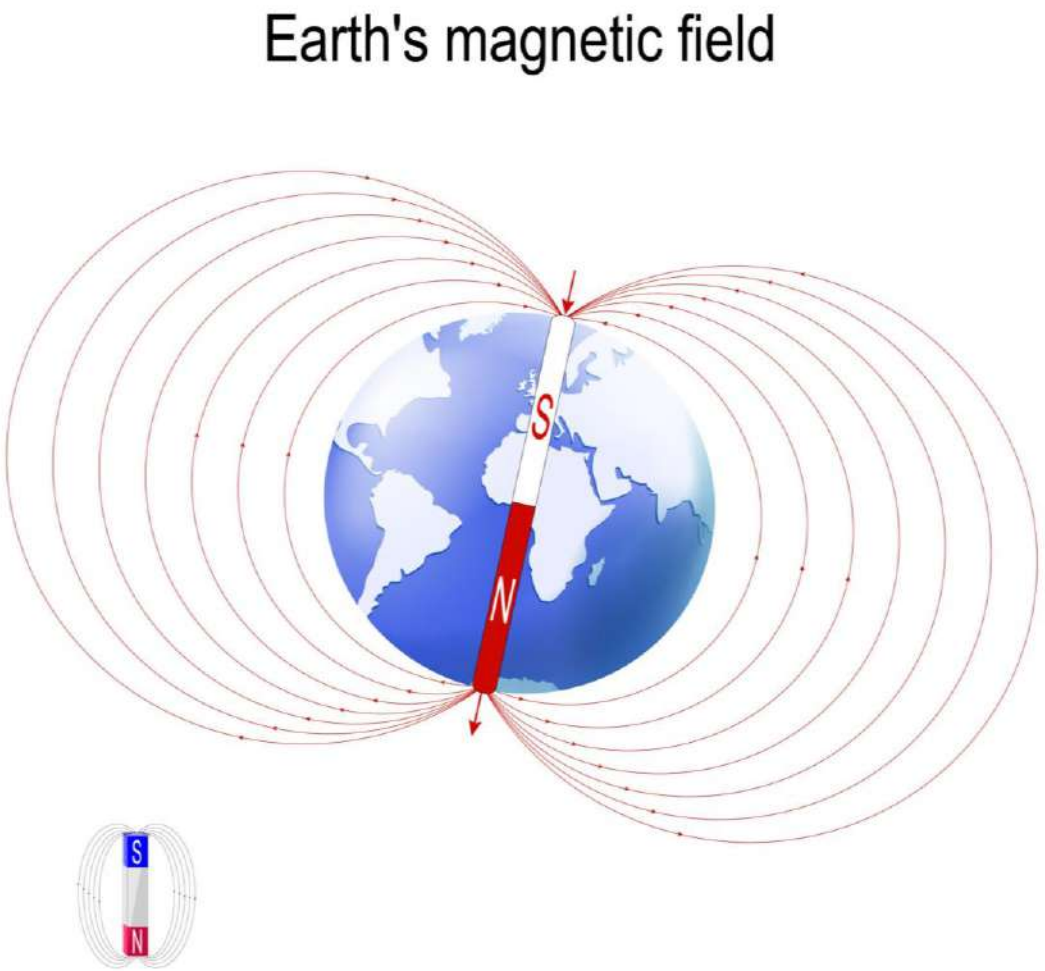
- **Vector Quantity:** It possesses both magnitude (strength) and direction.
- **Continuous Closed Loops:** Magnetic field lines emerge from the **North pole** and enter the **South pole** outside the magnet. Inside the magnet, they travel from the **South pole to the North pole**, forming continuous unbroken loops.
- **Never Intersect:** Two magnetic field lines never cross each other. If they did, it would mean a compass needle placed at the intersection point would point in two different directions at once, which is physically impossible.
- **Relative Strength via Density:** The degree of closeness (density) of the field lines indicates the strength of the magnetic field. The field is strongest near the poles, where the lines are packed tightly together.

Magnetic field



Magnetic field of

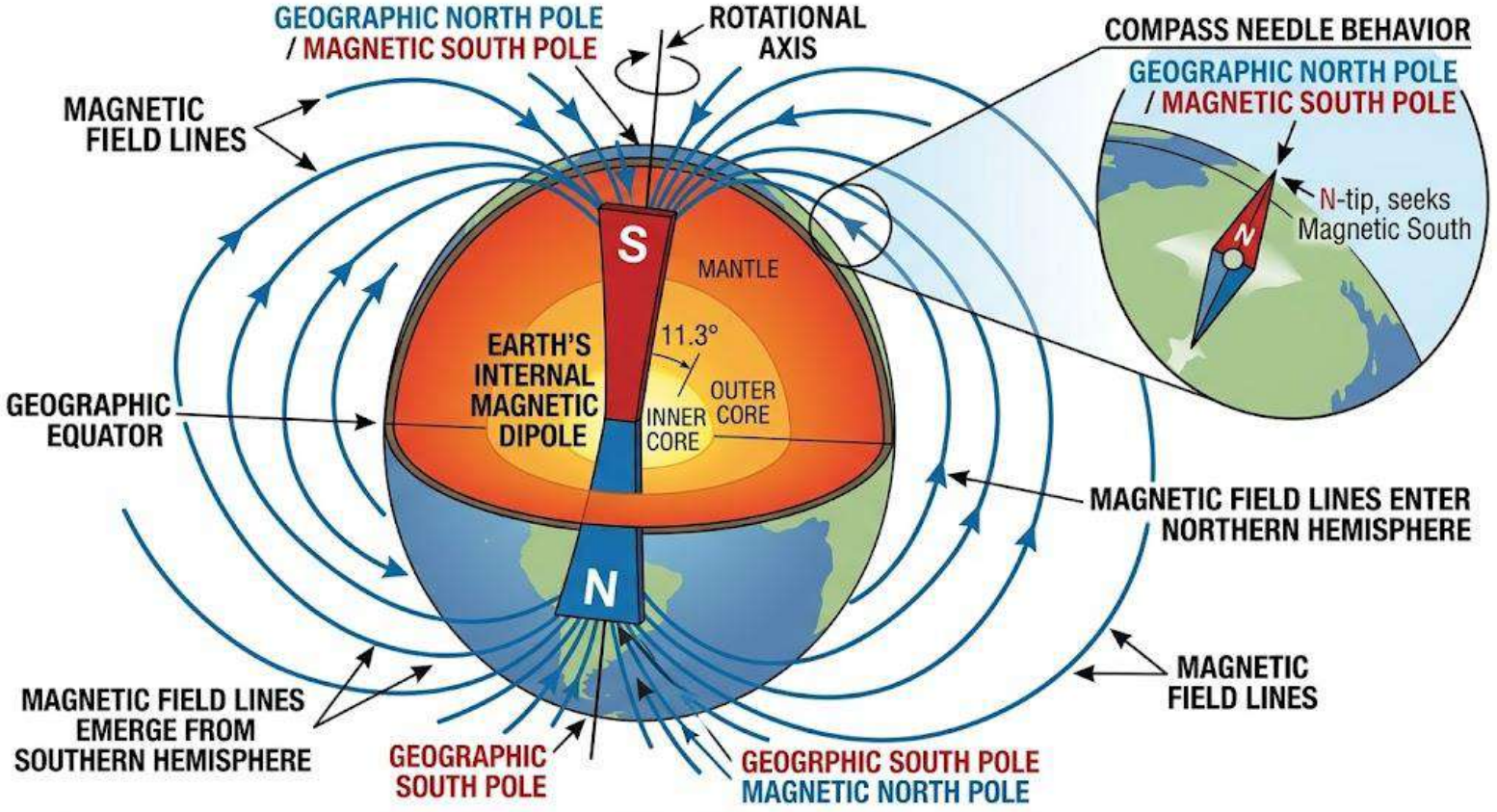
Earth behaves like a giant bar magnet tilted at an angle of roughly 11.3°relative to its rotational axis. This magnetic field extends thousands of kilometres into space, forming a protective bubble called the **magnetosphere**.



How Earth's magnetic field works.

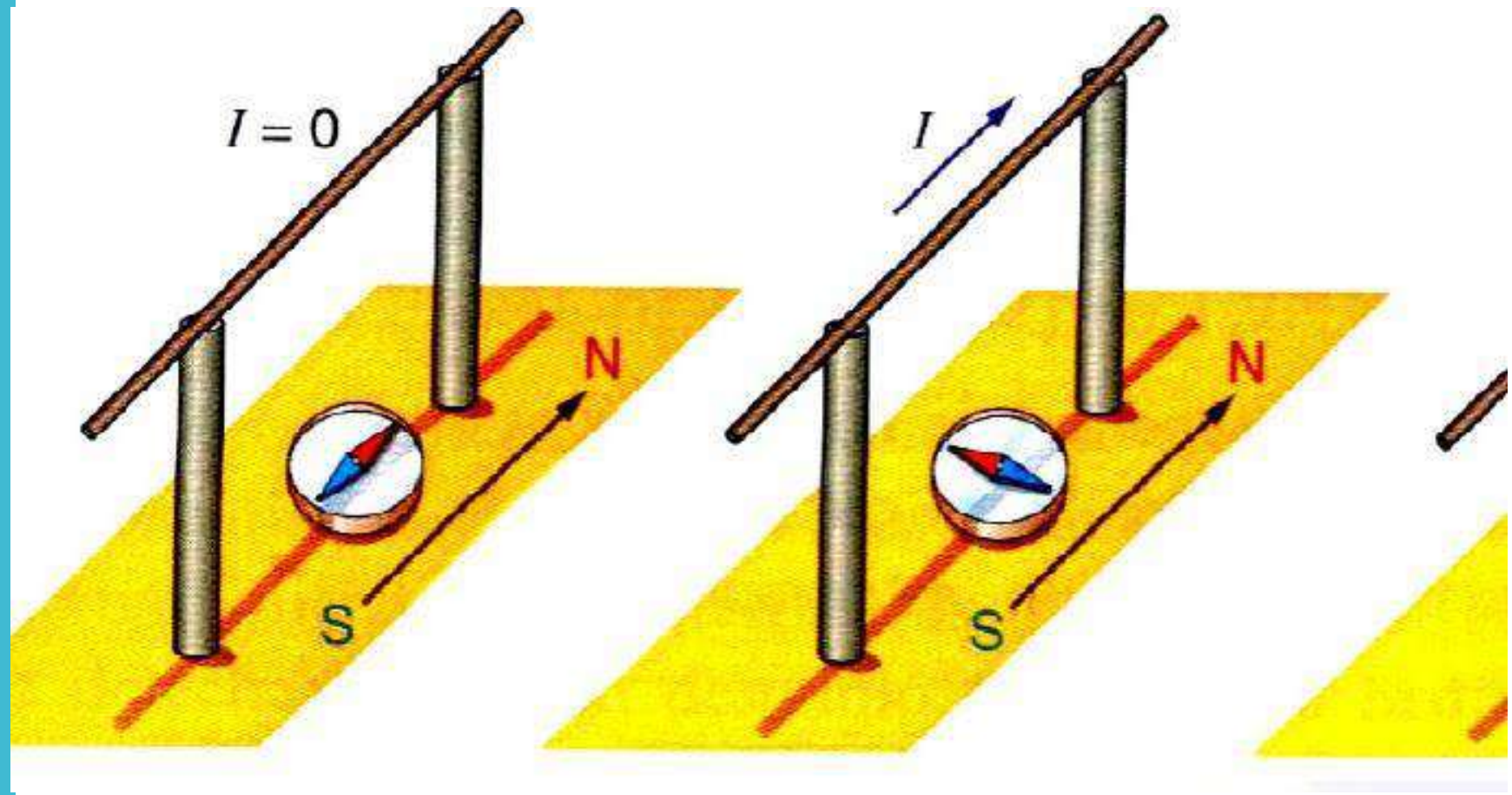
- **The Dynamo Effect (Origin):** Earth's magnetic field is generated deep inside its outer core. The heat from the inner core causes molten iron and nickel to move in convection currents. Combined with Earth's rotation, these moving conductive metals generate electric currents, which produce the magnetic field.
- **Magnetic Poles vs. Geographic Poles:**
  - Earth's **Geographic North Pole** is near the **Magnetic South Pole** (physically attracts north-pointing compass needles).
  - Earth's **Geographic South Pole** is near the **Magnetic North Pole**.
  - The magnetic field lines emerge near the Southern Hemisphere and curve around to enter near the Northern Hemisphere.

How Earth's magnetic field works.

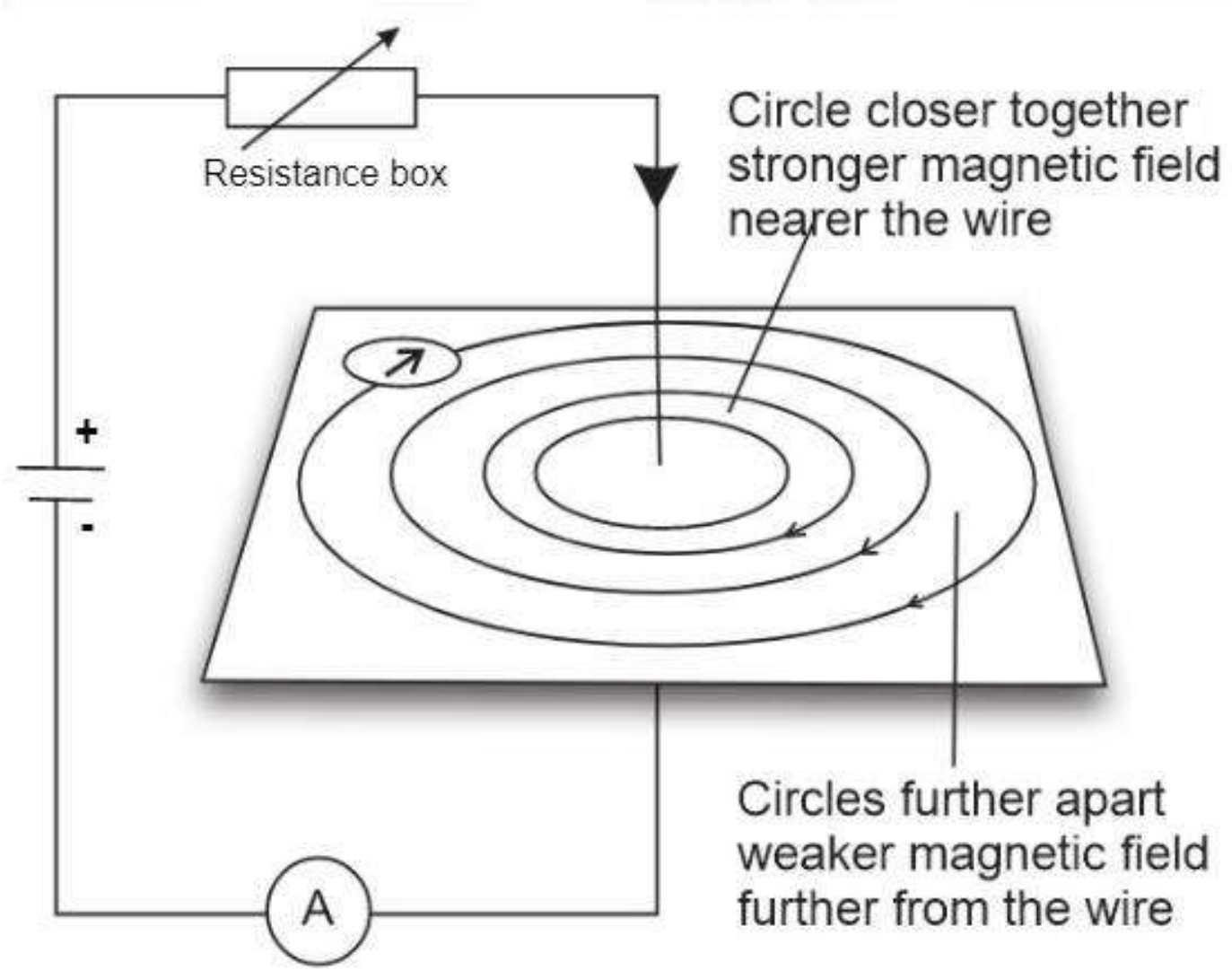


Magnetic Effect of Current (or Electromagnetism)

A current flowing in the wire always gives rise to a magnetic field around it.



Magnetic Field due to a straight current-carrying conductor

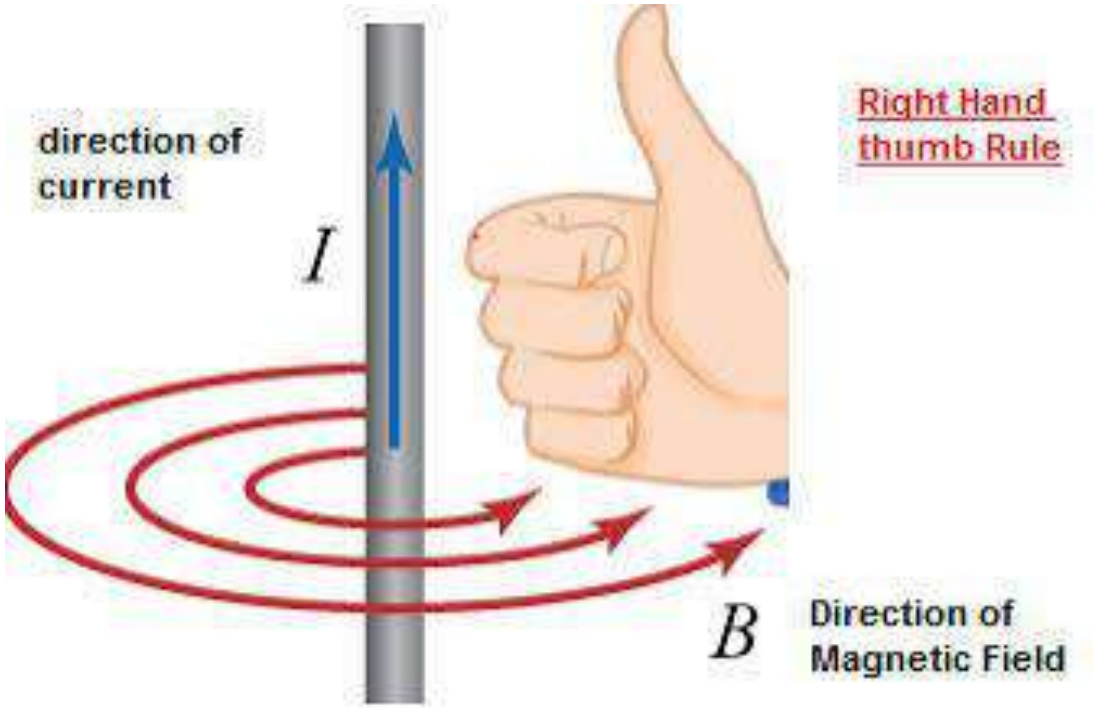


Magnetic Field due to a straight current-carrying conductor

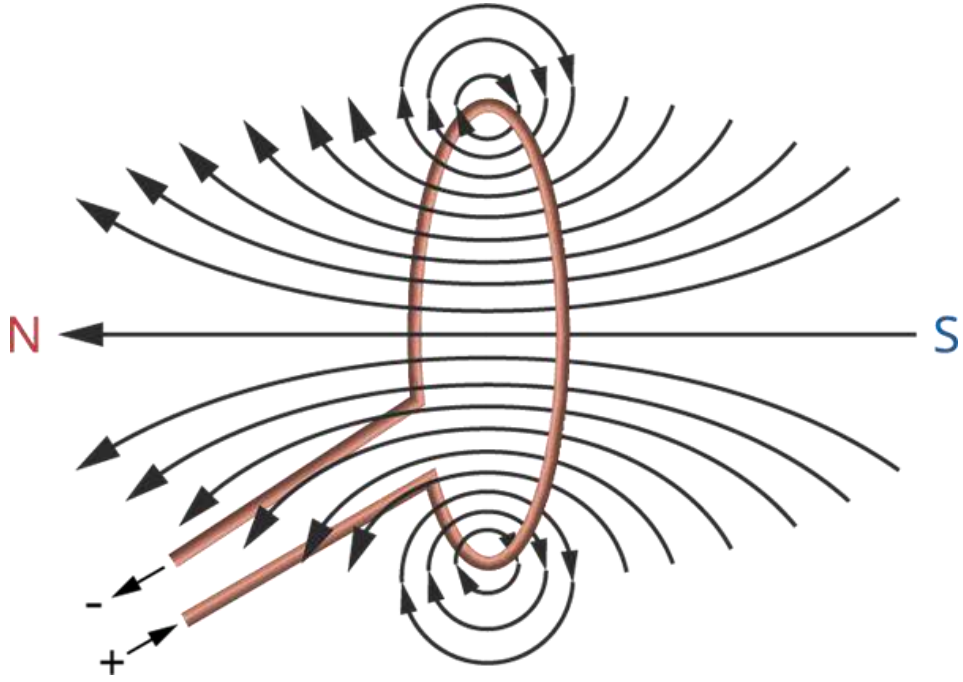
- Pattern: The magnetic field lines form concentric circles around the wire, centred on the conductor itself.
- Plane of Field Lines: The circular field lines lie in a plane perpendicular to the straight wire.
- Field Strength (B):
  - Proportional to Current : Increasing the electric current (I) strengthens the magnetic field ($B \propto I$).
  - Inversely Proportional to Distance: Moving further away from the wire (r) weakens the magnetic field ($B \propto \frac{1}{r}$). The field lines become farther apart as distance increases.

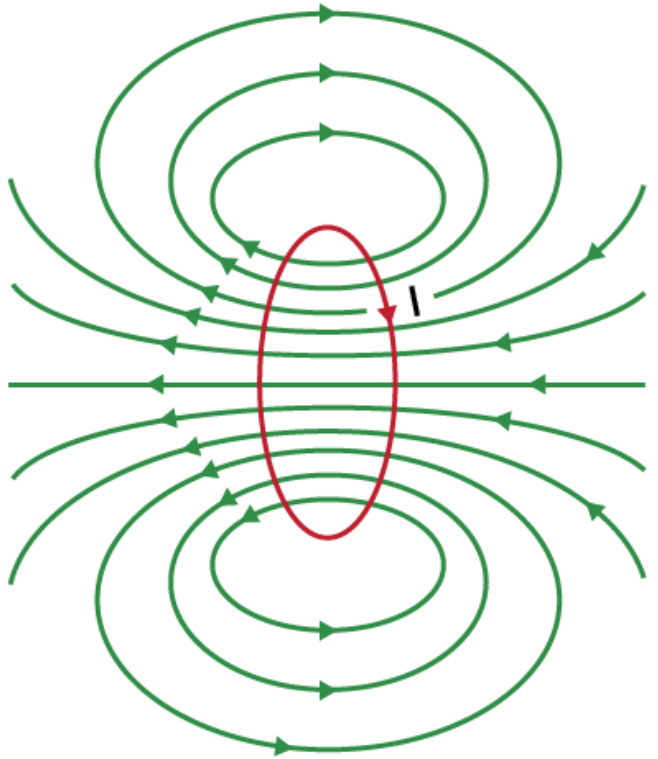
Right-hand Thumb rule

Imagine holding the straight conductor in your **right hand** such that your thumb points in the direction of the **electric current**. The direction in which your fingers curl around the wire gives the direction of the **magnetic field lines**.



Magnetic field pattern due to a circular loop carrying current.

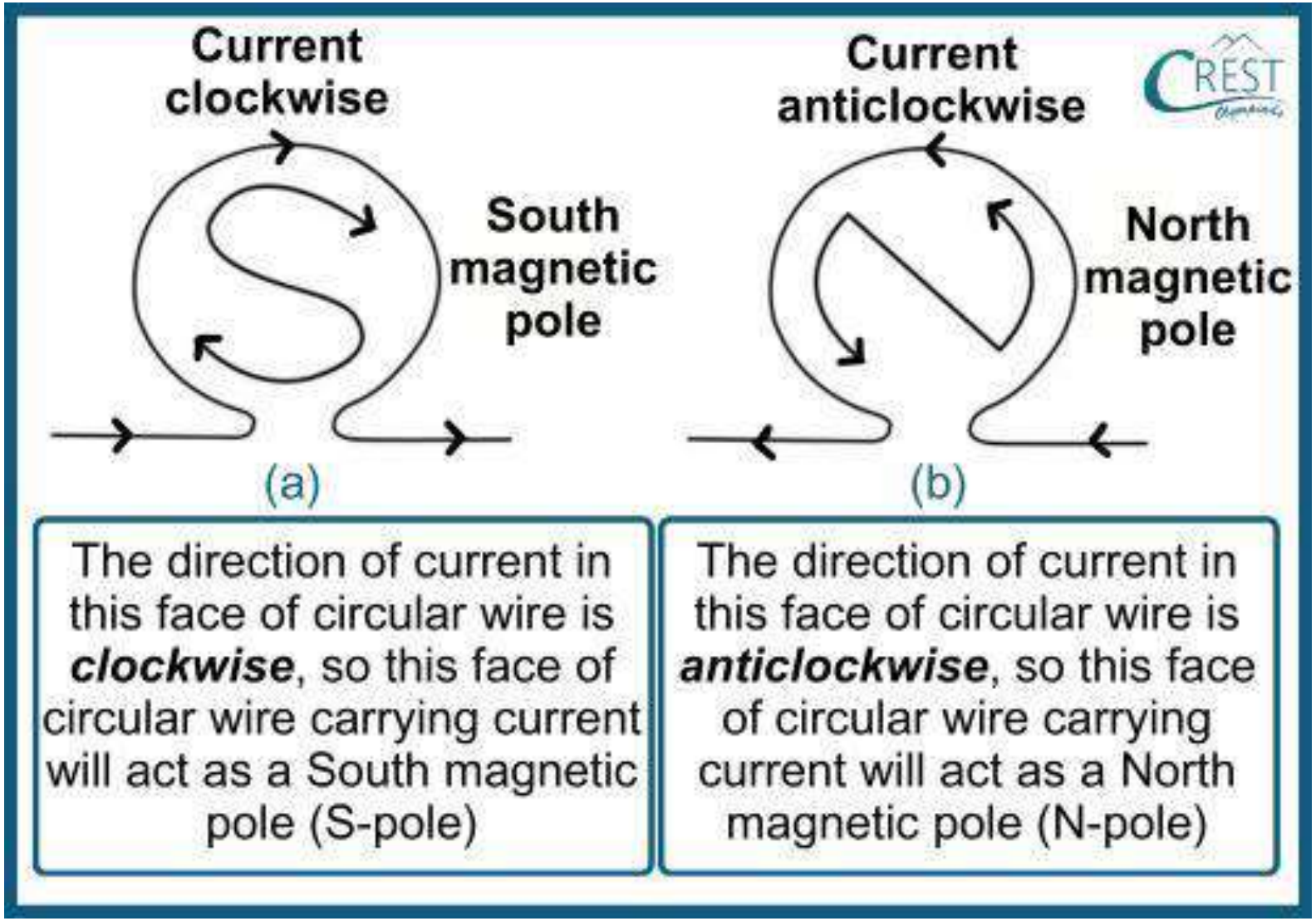




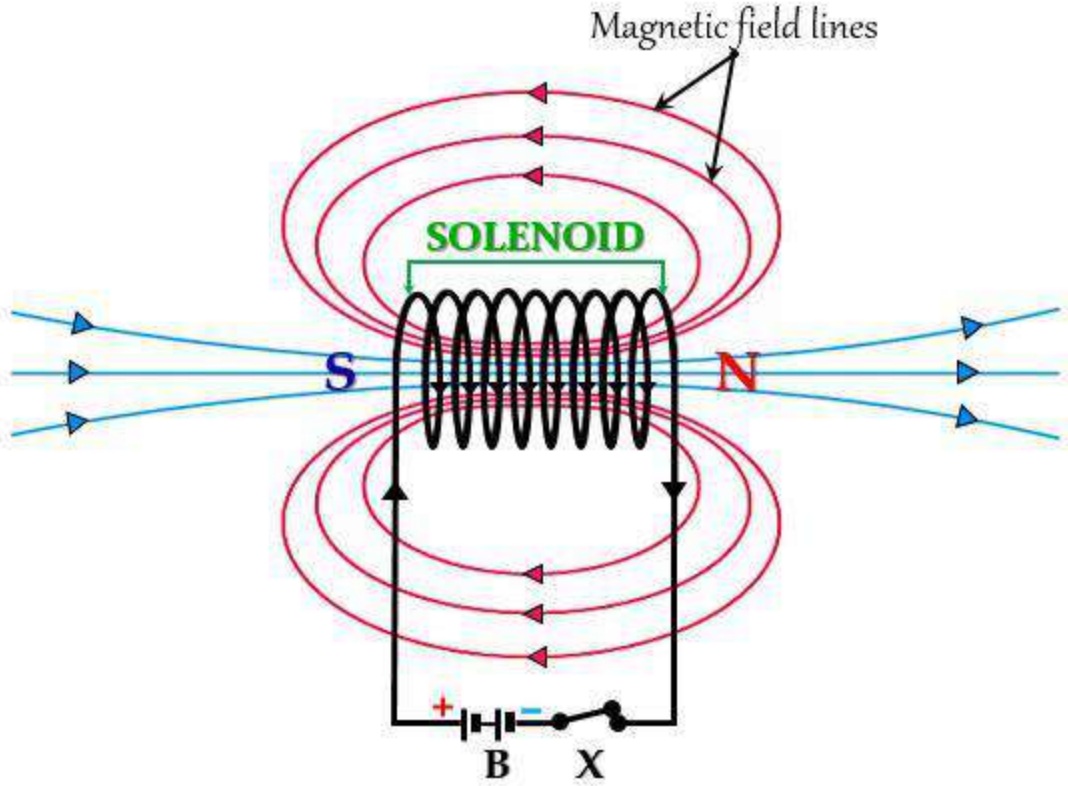
Magnetic field pattern due to a circular loop carrying current.

- At **every small portion of the wire**, the current produces a magnetic field around that portion. When you add all these small magnetic fields together:
- At the **centre of the loop**, the field contributions point in the **same direction**.
- They therefore reinforce each other.
- The resulting field passes through the centre approximately along the **axis of the loop**.
- So the circular loop produces a magnetic field pattern similar to a **bar magnet**.

Clock-Face Rule



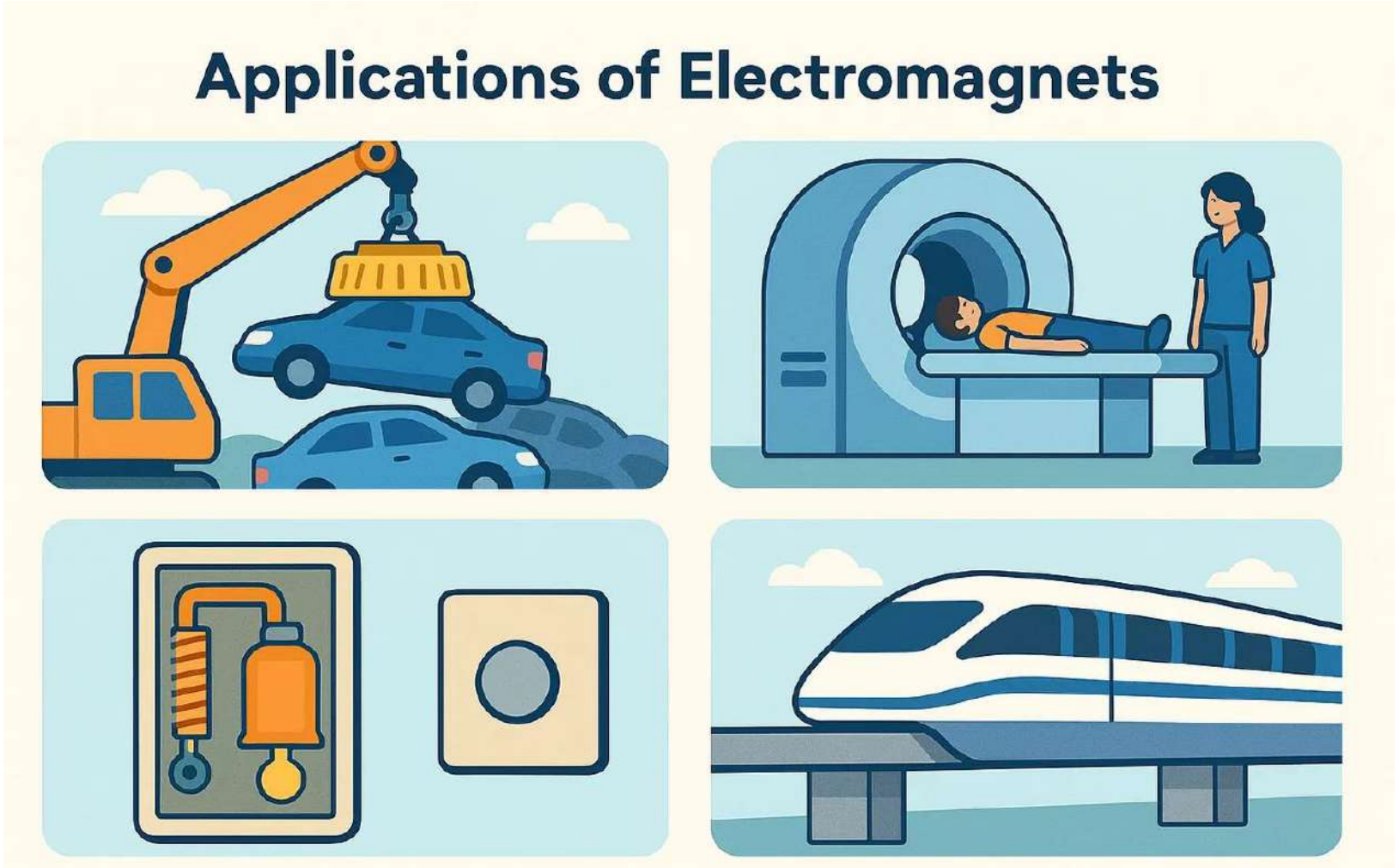
Magnetic Field due to a current carrying solenoid



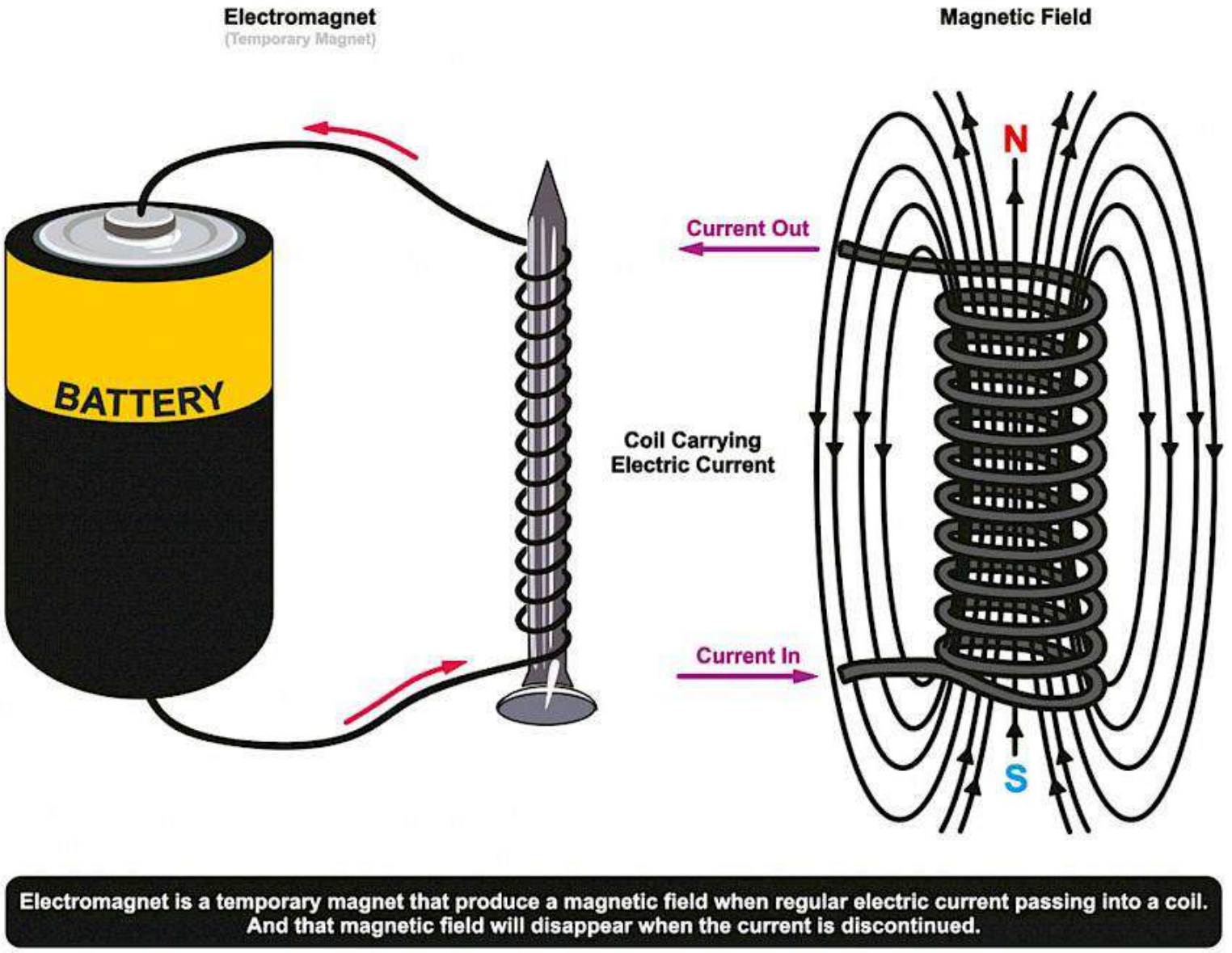
Magnetic field due to solenoid.

- A solenoid is a long coil of insulated wire wound tightly into a cylindrical helical shape, resembling a metallic spring. When an electric current flows through this coil, it generates a magnetic field around and inside it.
- **Inside the Solenoid:** The magnetic field lines combine into straight, parallel lines running close together along the length of the cylinder. This means the magnetic field inside is **uniform and very strong** everywhere.
- **Outside the Solenoid:** The magnetic field lines spread out and loop around from one end to the other, making the field **weak and non-uniform** on the exterior.
- **Behavior like a Bar Magnet:** One end of the solenoid acts as a **North Pole** (where field lines emerge) and the other end acts as a **South Pole** (where field lines enter). The overall magnetic pattern is identical to that of a standard bar magnet.

Electromagnet



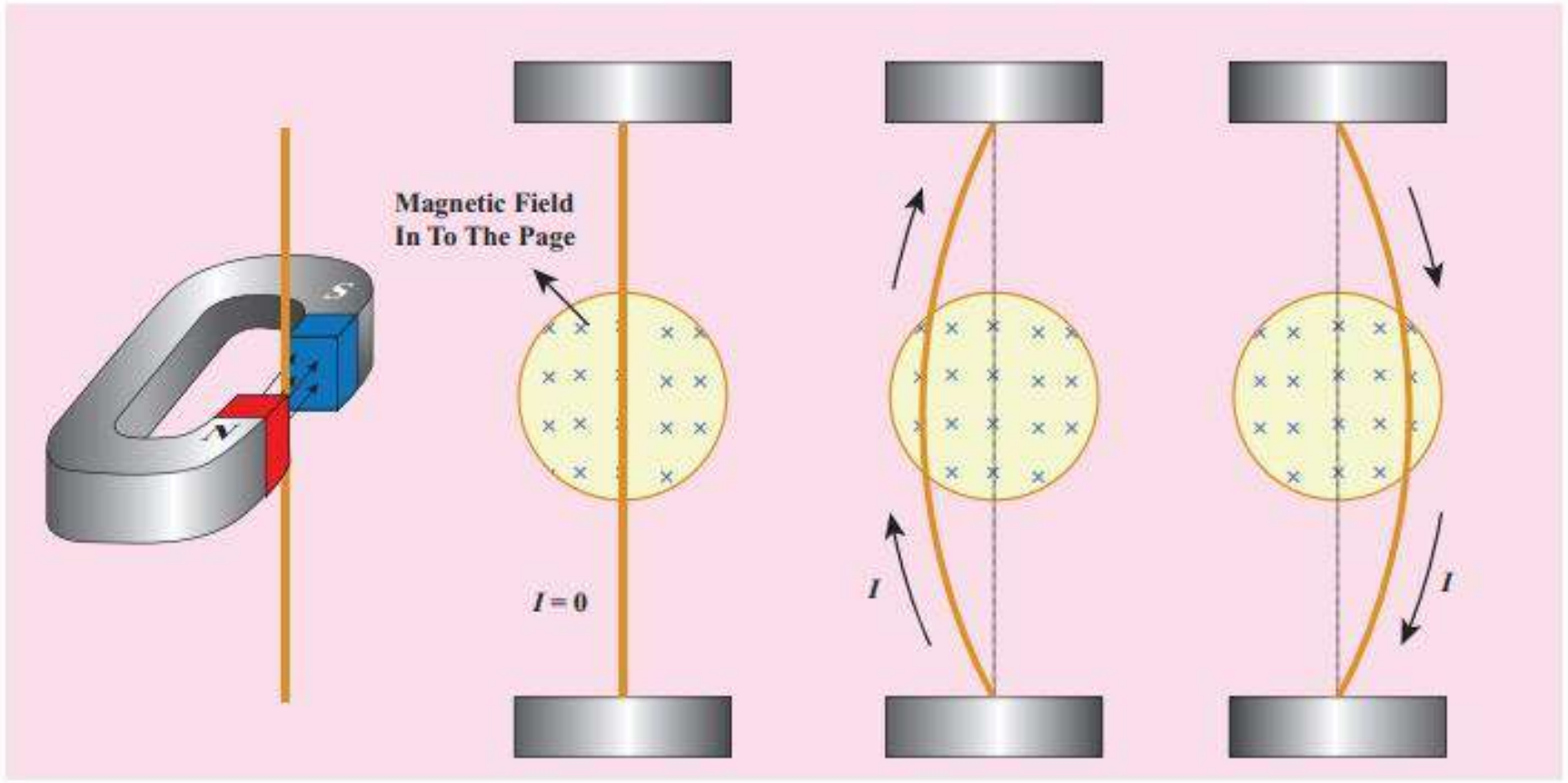
Electromagnet



Factors affecting the strength of an electromagnet

Factor	To Make Electromagnet Stronger	To Make Electromagnet Weaker
Current	Increase voltage / current	Decrease current
Coil Turns	Add more loops of wire	Remove loops of wire
Core Material	Use soft iron	Remove core or use plastic/wood
Coil Density	Pack turns closer together	Spread turns far apart

Force on a current-carrying conductor in a magnetic field

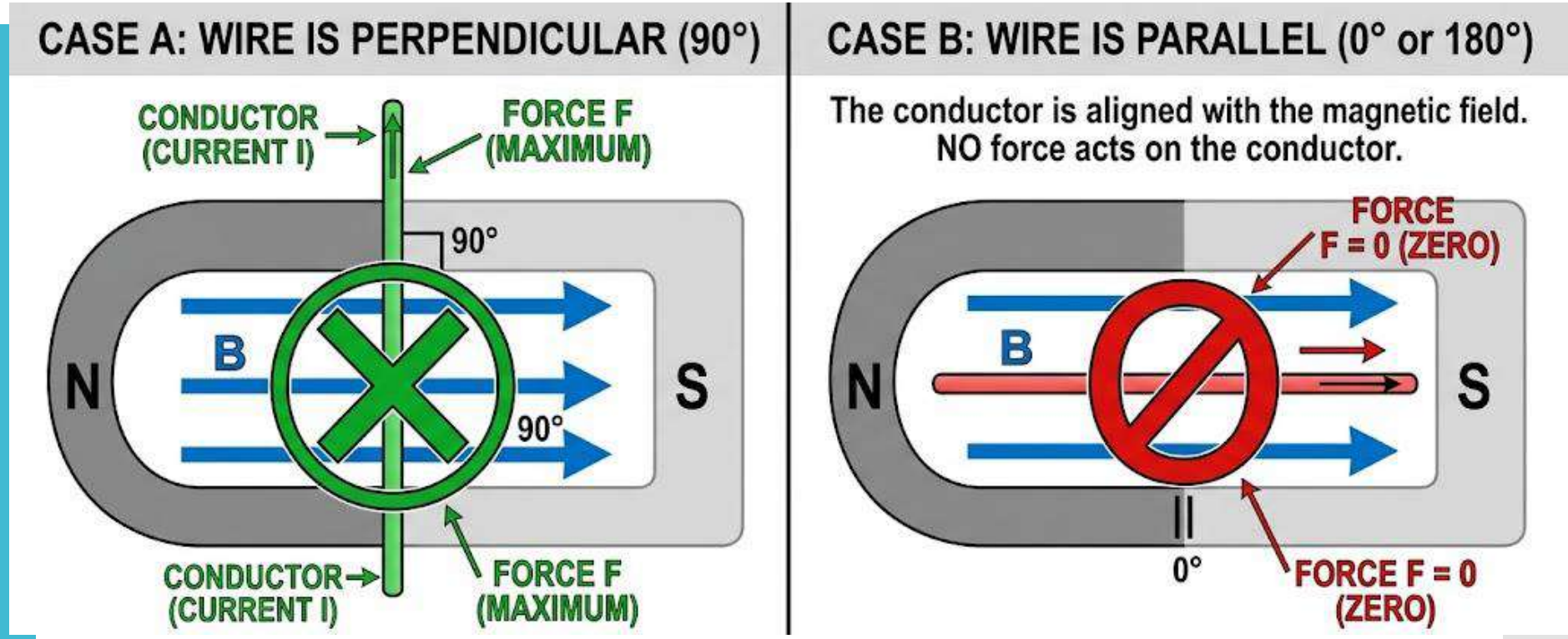


Force on a current-carrying conductor in a magnetic field

- When a current-carrying wire is placed inside an external magnetic field, it experiences a **physical push or force**.
- The magnitude of this force depends on three main things:

1. **Amount of Current:** Passing more current through the wire increases the force.
2. **Strength of External Magnet:** A stronger magnet exerts a greater force.
3. **Length of Conductor inside Field:** A longer wire segment inside the magnetic field experiences more force.

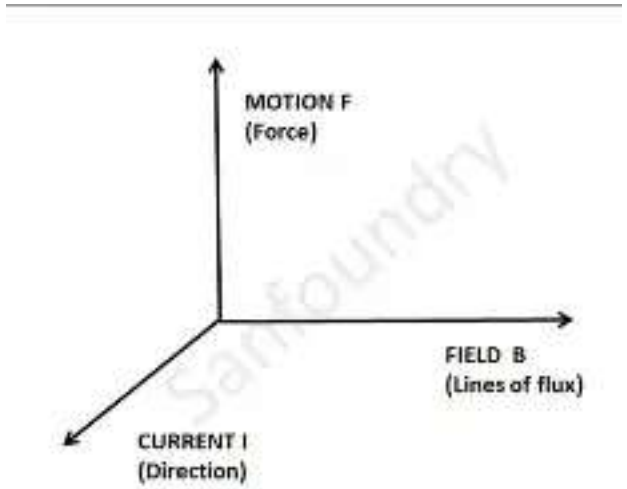
Force on a current-carrying conductor in a magnetic field



The force is maximum when the wire is placed perpendicular (at 90°) to the magnetic field lines.
The force becomes zero if the wire is placed parallel to the magnetic field lines.

Fleming's left-hand Rule for the direction of force on a current-carrying conductor.



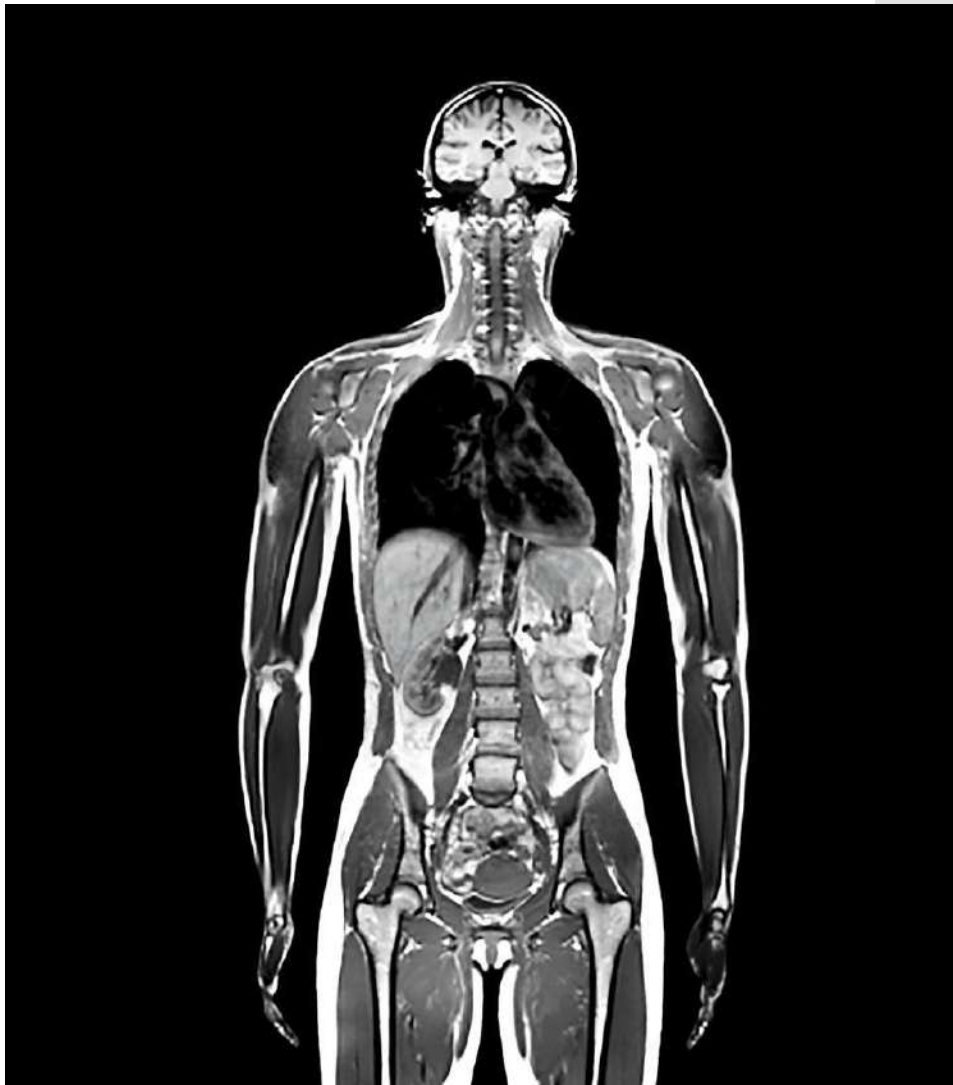


Fleming's left-hand Rule for the direction of force on a current-carrying conductor.

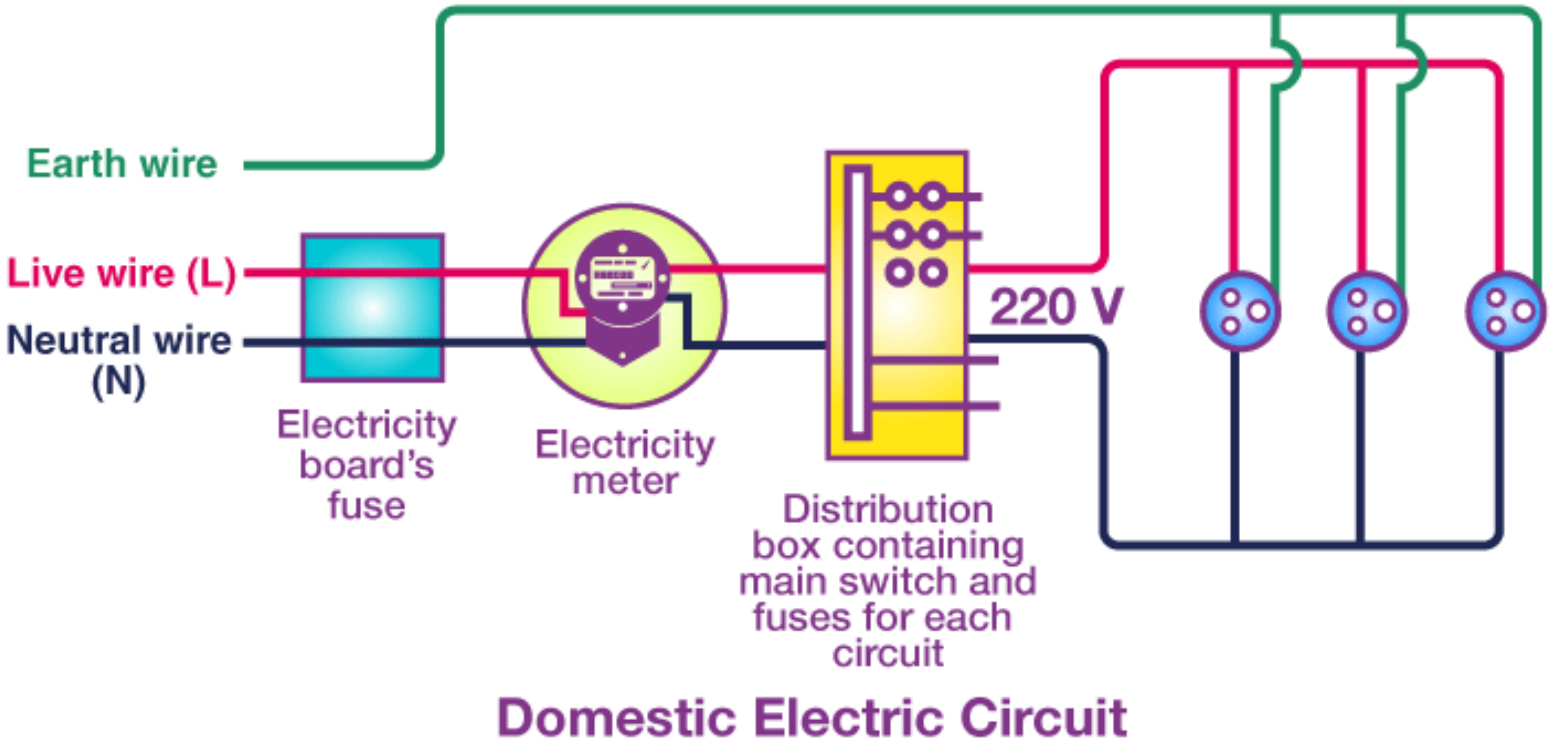
- If the thumb, forefinger and middle finger of the left hand are held mutually perpendicular to each other, with the forefinger pointing in the direction of the magnetic field and the middle finger in the direction of current, then the thumb gives the direction of force (or motion) on the conductor.

Magnetic Resonance Imaging





Domestic Electric circuits



Domestic Electric Circuit

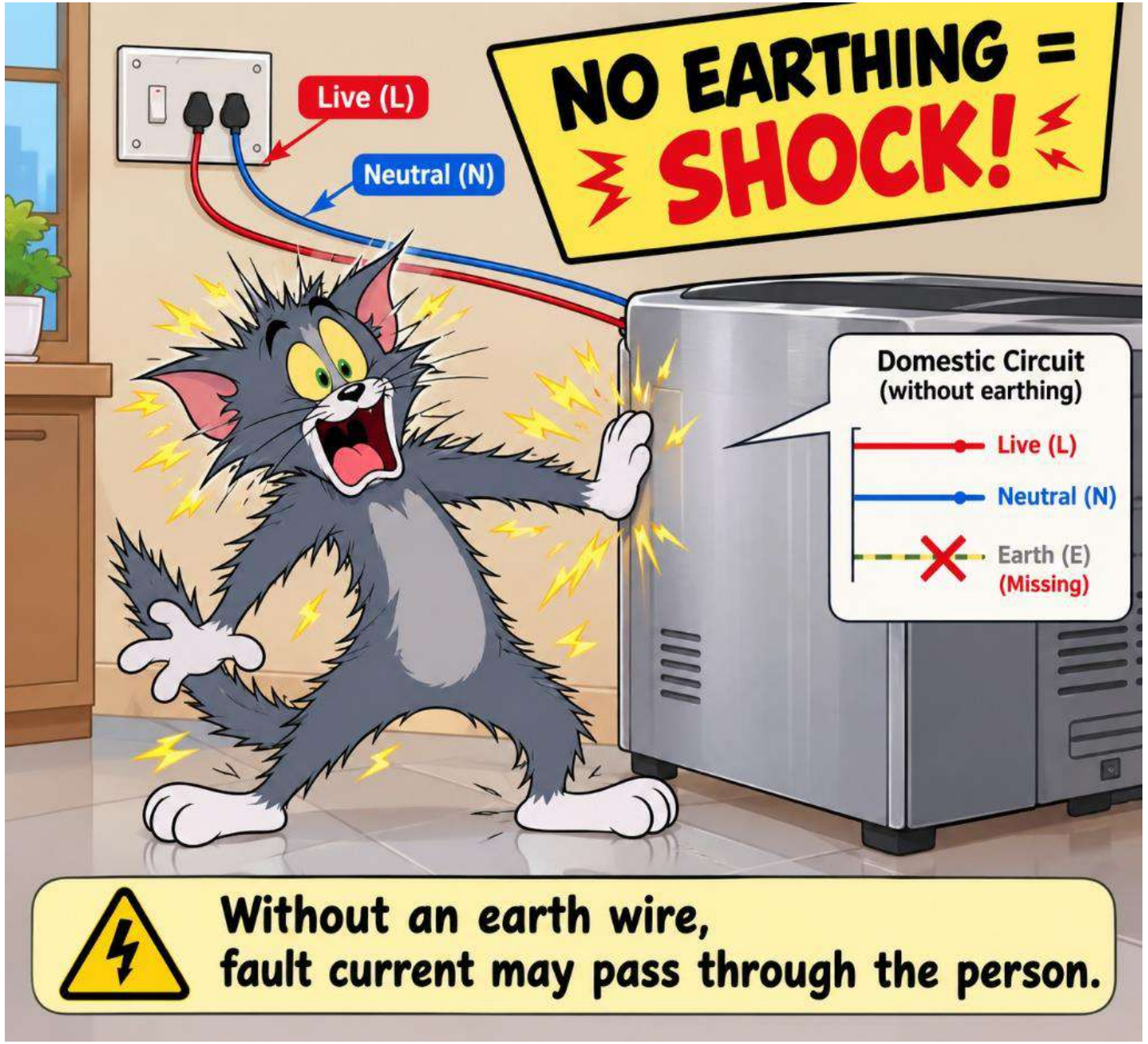


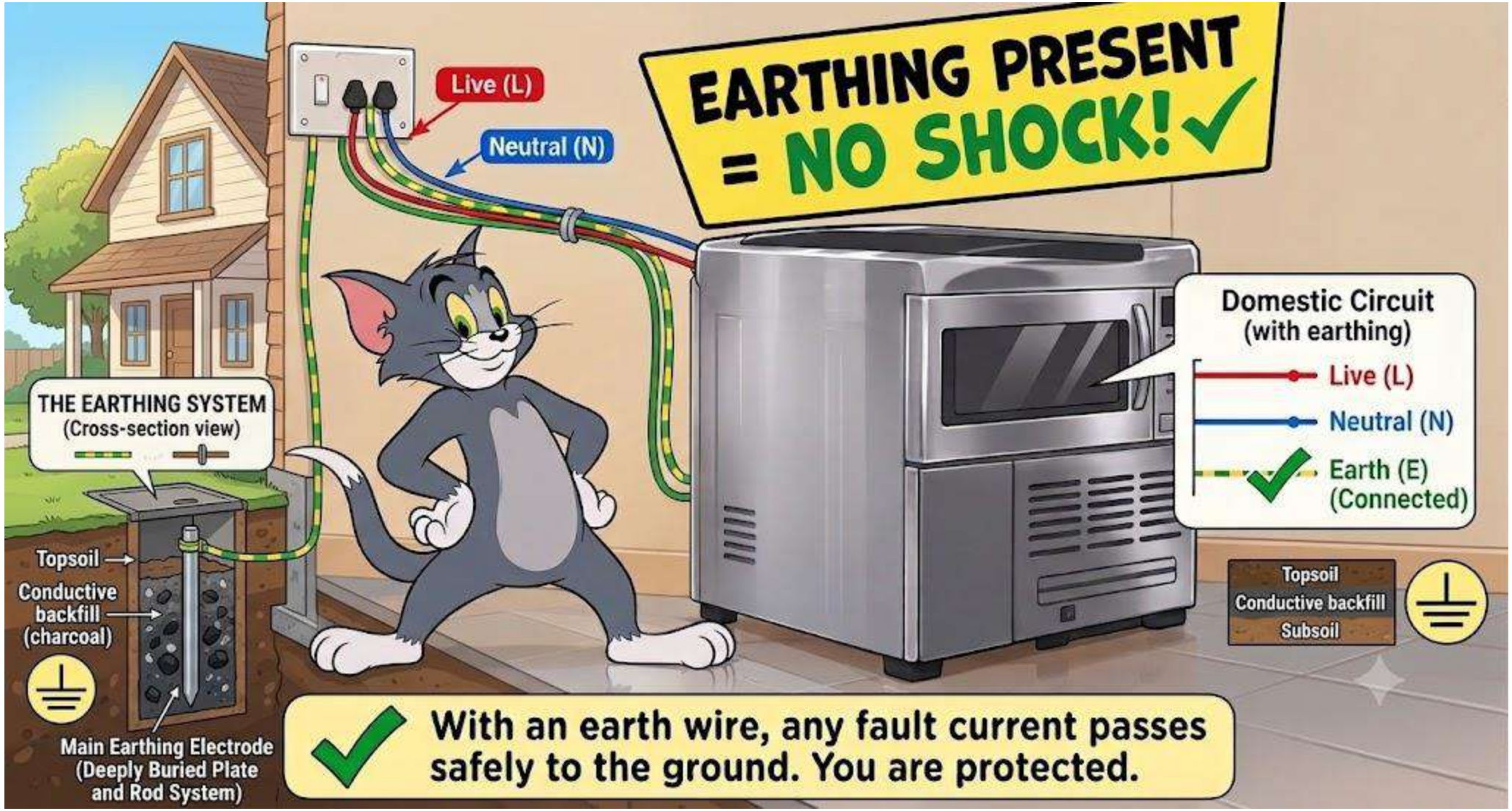
Domestic circuit board rated for 5A current used for low power appliances like fans, bulbs, mobile charging points, etc.



Domestic circuit board rated for 15 A current used for high-power appliances like a washing machine, geyser, air conditioners, etc

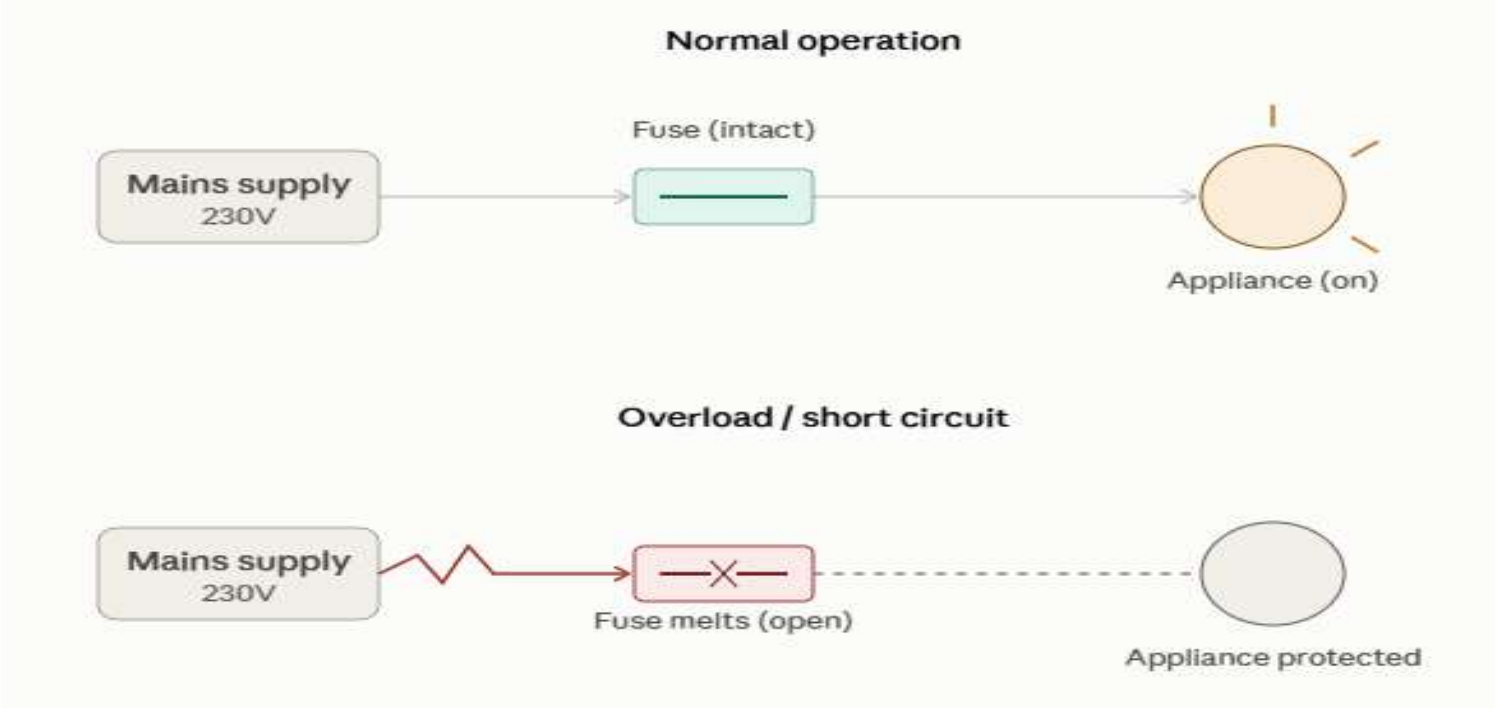
Earth Wire


NO EARTHING = SHOCK!
Live (L)
Neutral (N)
Domestic Circuit
(without earthing)
Live (L)
Neutral (N)
Earth (E)
(Missing)
Without an earth wire,
fault current may pass through the person.


EARTHING PRESENT
= NO SHOCK!
Live (L)
Neutral (N)
Domestic Circuit
(with earthing)
Live (L)
Neutral (N)
Earth (E)
(Connected)
THE EARTHING SYSTEM
(Cross-section view)
Topsoil
Conductive backfill (charcoal)
Main Earthing Electrode
(Deeply Buried Plate and Rod System)
Topsoil
Conductive backfill
Subsoil
With an earth wire, any fault current passes safely to the ground. You are protected.

Overloading in Domestic circuits.



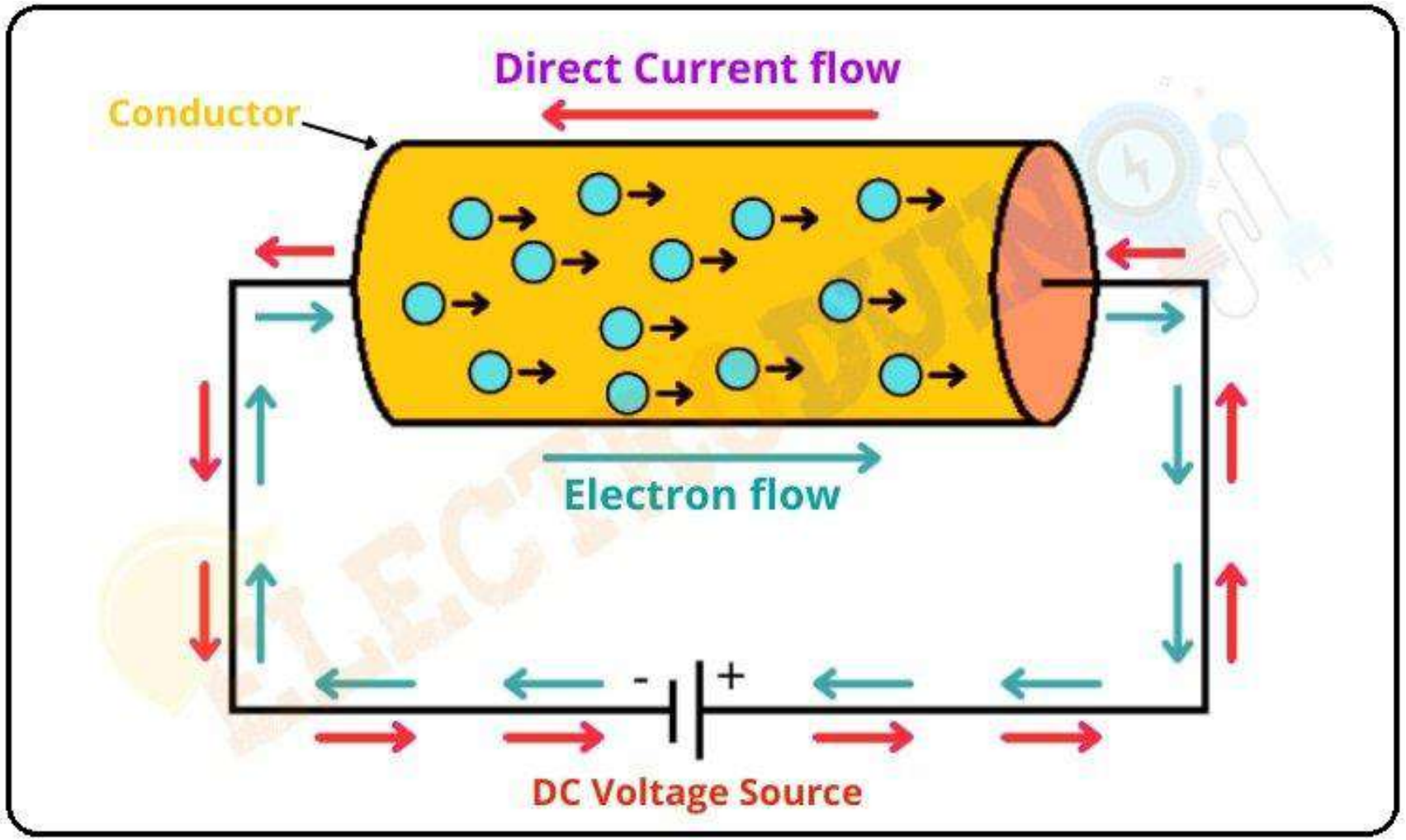


Electric fuse

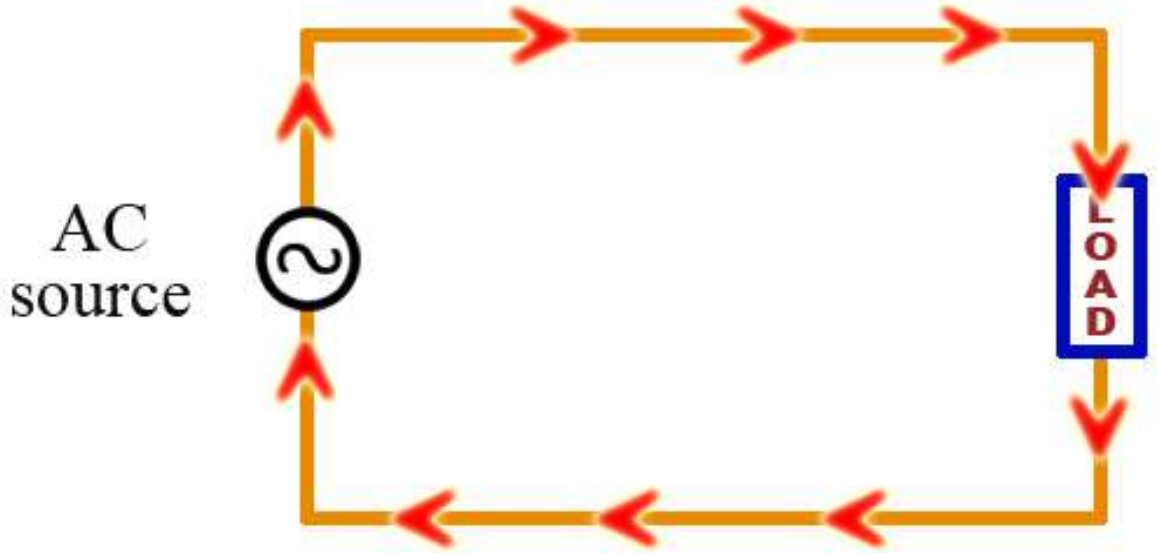


An electric fuse is a safety device designed to protect electrical circuits from overloading and short circuits. It works by melting a thin metal wire when excess current flows through it, instantly breaking the circuit to prevent damage and fire hazards.

Direct Current



Alternating current

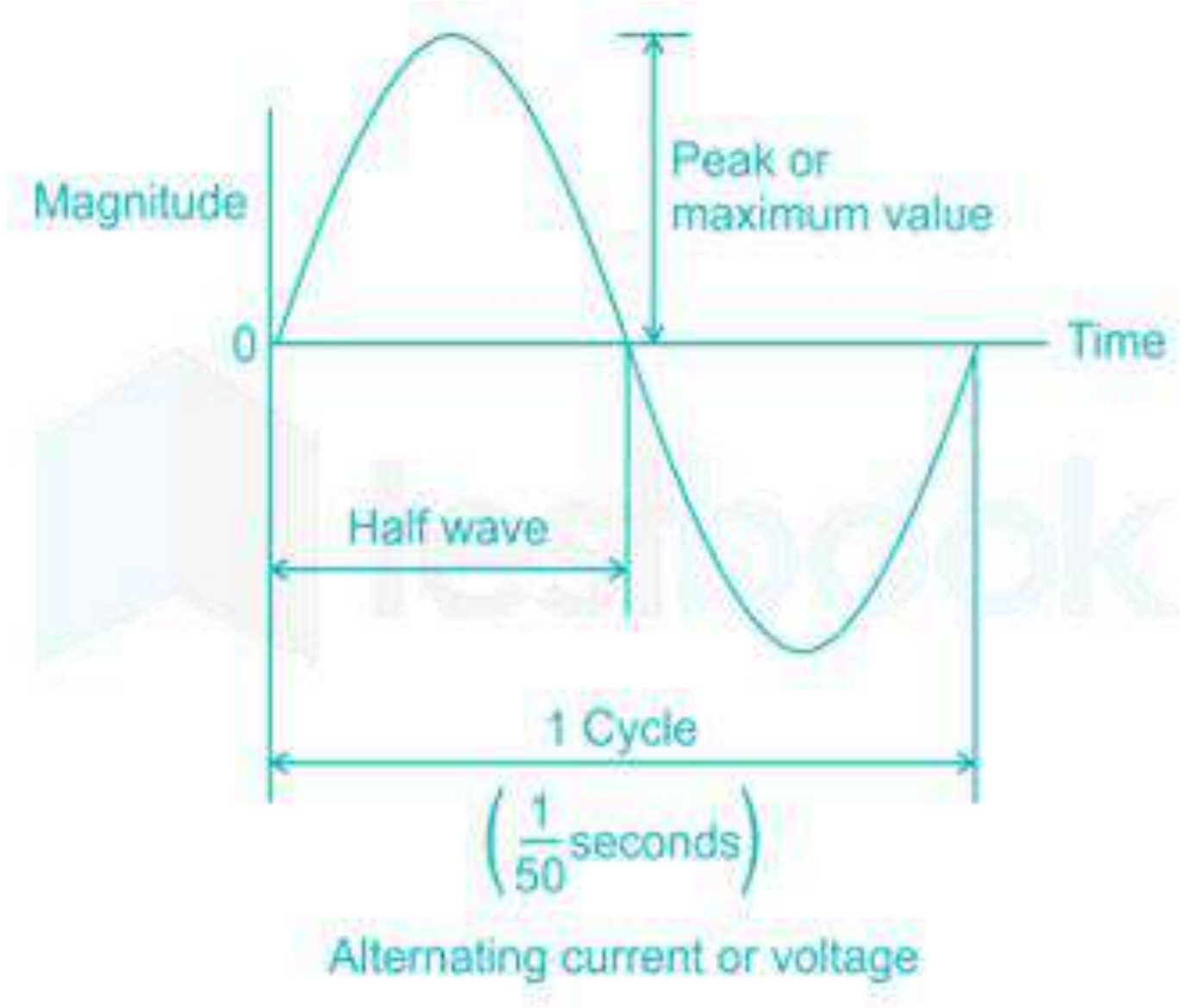


Alternating Current

AC and DC



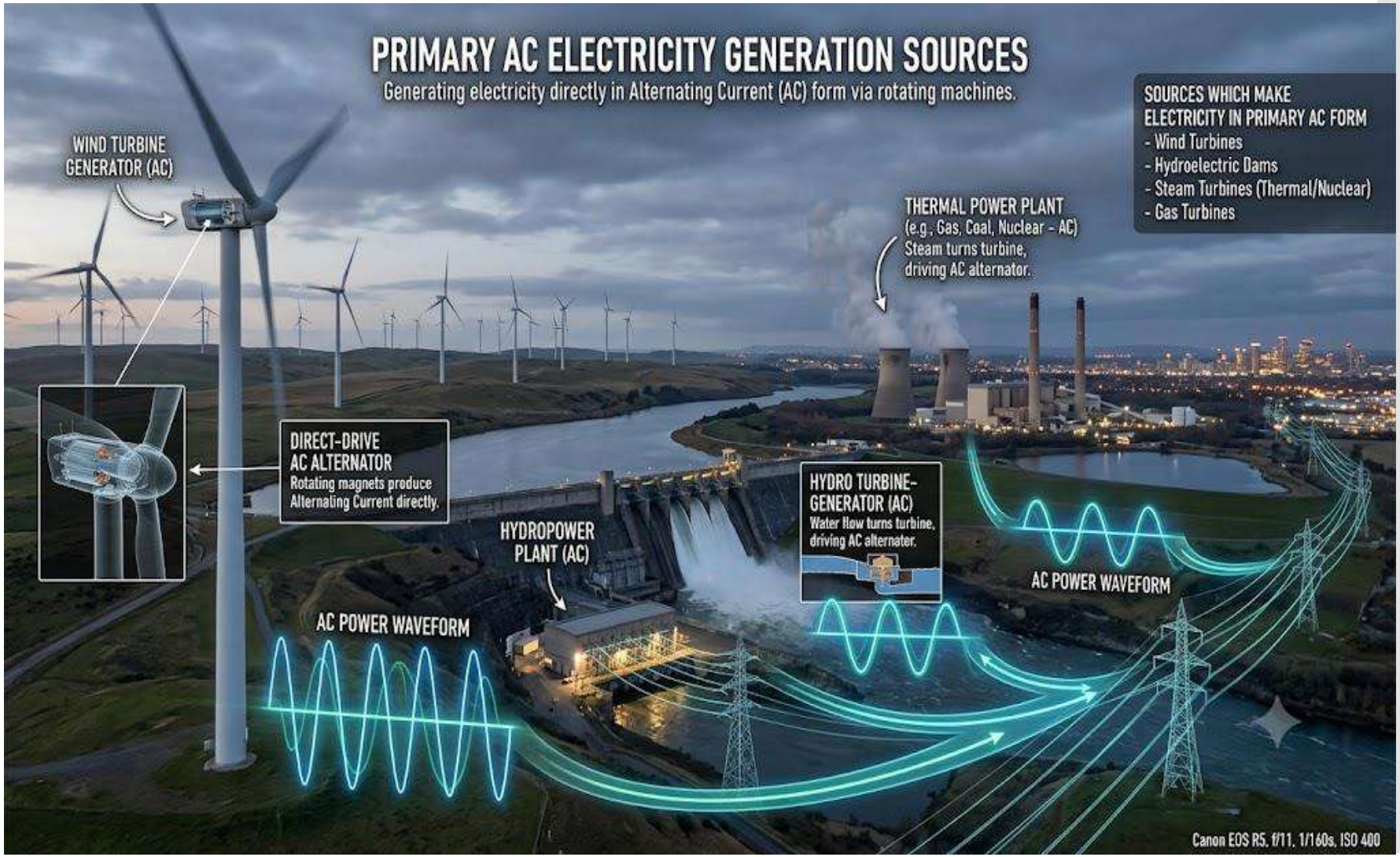
Alternating current in India



DC current Sources



AC Current Sources



Direct Current requiring household Devices



Alternating Current requiring household devices



Why Alternating current is used in Household.

- **Easier and Cheaper Bulk Generation:** AC is mechanically simpler and less expensive to generate on a large scale.
- **Efficient Voltage Stepping via Transformers:** AC can be stepped up to very high voltages for long-distance transport and stepped down to safe household levels using simple, passive transformers that operate at efficiency levels above 98%.
- **Minimal Power Loss During Transmission:** By stepping up AC to high voltages for grid transmission, the current *I* is minimised.

Why Alternating current is used in Household.

- **Simpler AC-to-DC Conversion (Rectification vs. Inversion):** Converting AC to DC at the consumer level requires inexpensive and reliable components like simple diode bridge rectifiers. Conversely, converting DC to AC requires complex, costly electronic inverters with active high-speed switches and control circuits.
- **Durable and Low-Maintenance AC Motors:** Household appliances like ceiling fans, washing machines, and refrigerators use AC induction motors. These motors do not require brushes or commutators, making them cheaper to manufacture, highly reliable, and virtually maintenance-free compared to DC motors.

Precautions in the use of electricity (1)



Precautions in the use of electricity (2)

Avoid operating plugs and switches with wet hands, as water conducts electricity.



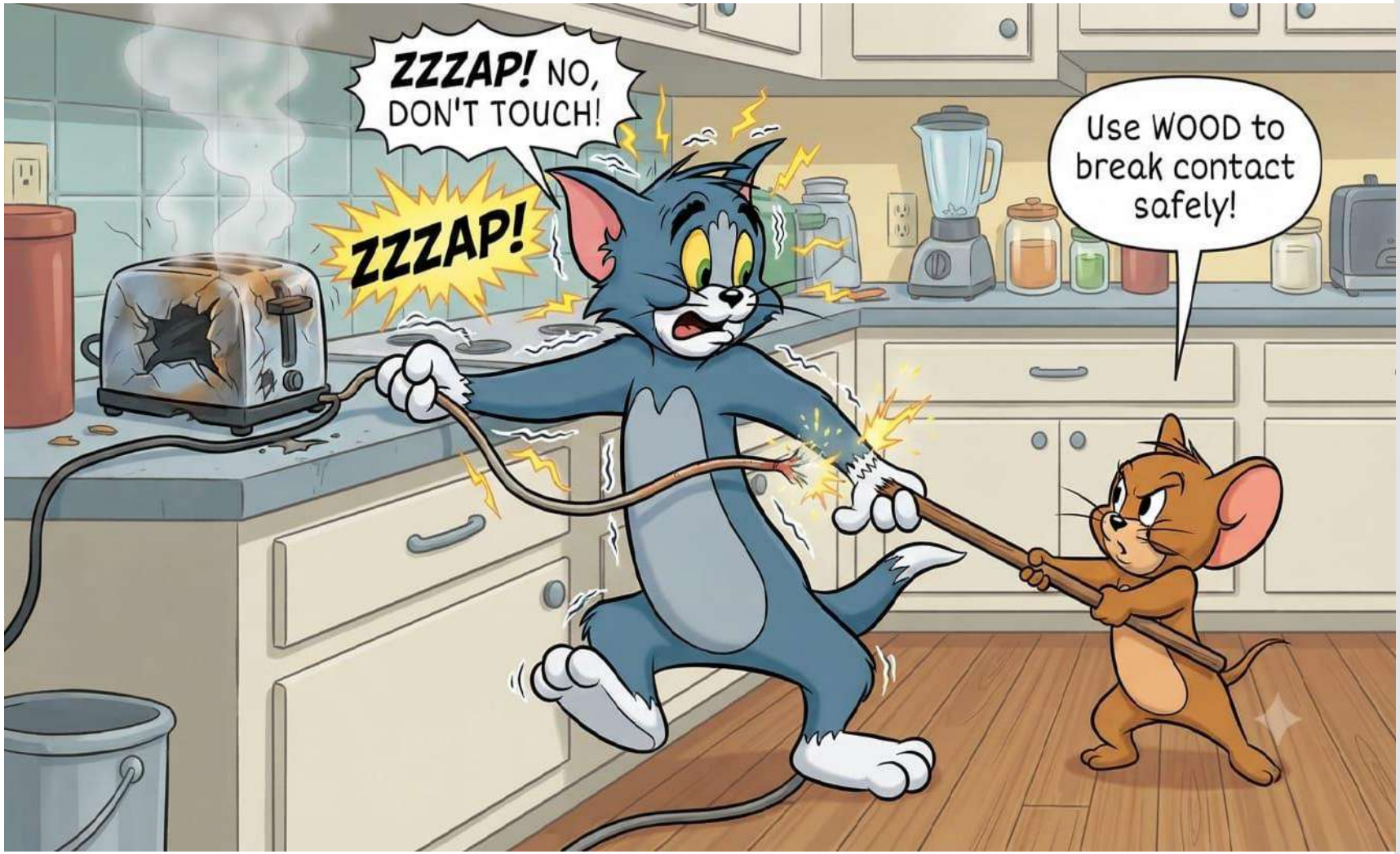
Precautions in the use of electricity (3)

High-power appliances like electric ovens, geysers, and air conditioners should be given earth connections



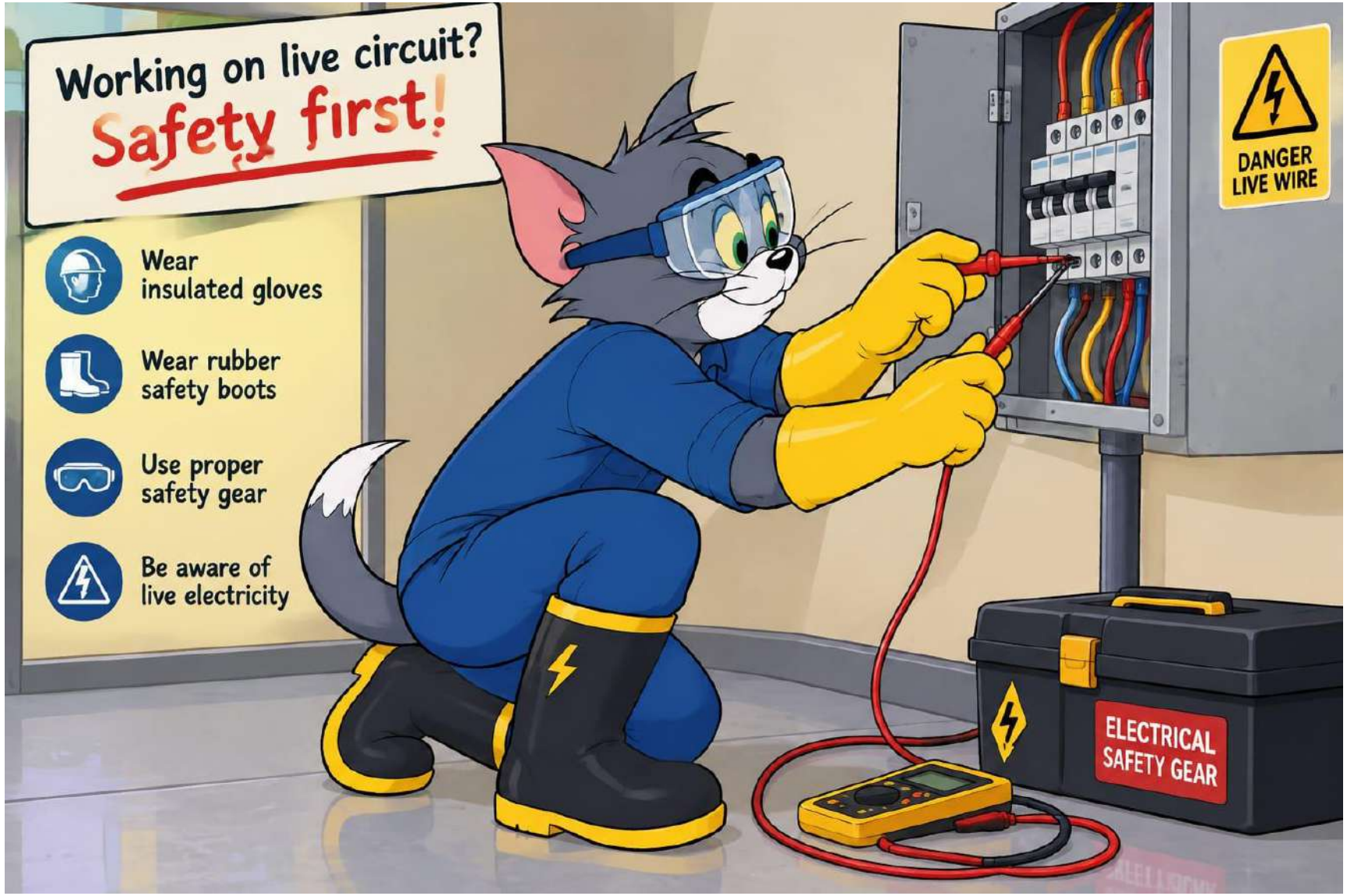
Precautions in the use of electricity (4)

The person who touches the live electric wire should be provided with an insulated support of wood, plastic, etc., using the insulator to separate the person from the live wire.



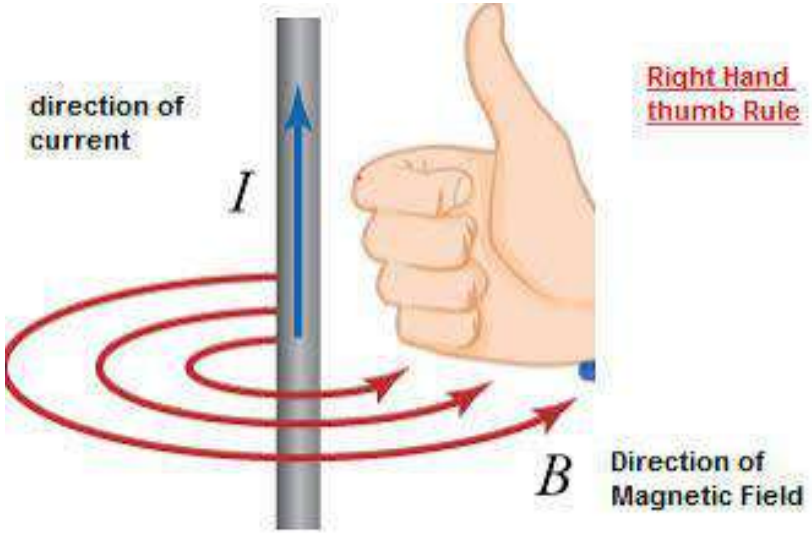
Precautions in the use of electricity (5)

If you must handle or work on a live circuit, protective gear like rubber gloves and rubber shoes must be put on.



Summary

- A current-carrying conductor creates a magnetic field around it.
- The right-hand thumb rule gives the direction of the magnetic field: the thumb points in the direction of the current, and the curled fingers point in the direction of the magnetic field



Summary

- When a current-carrying wire is placed inside an external magnetic field, it experiences a **physical push or force**.
- The direction of the experienced force is determined by Fleming's left-hand rule



Summary

- The domestic electric circuits contain three wires: the live wire, which has a voltage of 220V, the neutral wire at 0V and the earth wire for safety.

